Supplementary Appendix: Classification of NBER Working Papers Applying VSL

This appendix details the steps taken to classify NBER working papers that apply VSL. This exercise was used to assess whether VSL is typically applied to theoretically supported decisions or not when used in low and middle-income countries. The classification was done with Claude and may not perfectly replicate given randomization injected into large language models. This appendix was written in part by Claude, which I tasked with detailing the steps it took carefully to maximize replicability.

.1 Paper selection

I queried every paper published in the NBER working paper database since 2000 whose full text contained the term "value of a statistical life," "VSL", "'disability-adjusted life year," or "DALY." The search was run against EconLit's NBER working paper index in August 2026. The search returned 363 candidate papers. Each candidate was converted from PDF to Markdown (using `marker-pdf`), then the markdown was run through the two-stage classification procedure below.

.2 Two-stage classification

Stage 1 (screening). Every candidate paper was coded on two binary variables:

- `estimate_vsl`: does the paper's *own* empirical, structural, or theoretical analysis derive a new VSL estimate, rather than merely citing an existing one?
- `use_vsl`: does the paper take a VSL figure – its own or borrowed from the literature – and *apply* it, e.g. in a benefit-cost analysis, to evaluate a policy, or to compute a welfare effect? Computing a cost-per-DALY or DALYs-averted figure counts as an application; a literature-review mention with no application does not.

Papers coded `use_vsl` = 1 (240 of 363) proceeded to Stage 2.

Stage 2 (misapplication classification). Each `use_vsl` = 1 paper was further coded on:

- `developing_country`: VSL/DALYs are applied to a low- or middle-income population, or to a global/cross-country policy that includes such populations.
- `nonmarginal_risk`: VSL is applied to a decision that is not a marginal (small) change in mortality risk – e.g. a curative treatment, a large discrete intervention, or aggregation across a catastrophic event – rather than the small-risk-reduction context VSL was designed for.
- `transfer`: the paper prices a population's lives using that population's own (possibly income-adjusted) VSL, where that valuation feeds a decision actually financed or decided by a different, wealthier entity – e.g. a rich country, a multilateral donor, or a globally coordinated process dominated by rich countries.

- `transfer_correct`: the paper applies a funder's VSL to value benefits in a population it is externally financing.
- `self_aware`: relevant only when `nonmarginal_risk` or `transfer` equals 1; the authors explicitly flag the application as imperfect or a methodological stretch, but use it anyway for lack of a better alternative.
- `givewell`: the study evaluates an intervention financed, in whole or part, by GiveWell.

For papers with `developing_country` = 1, I additionally instructed Claude to code `transfer_category`, classifying the paper's subject matter as *health_ program* (vaccines, deworming, HIV/ART, bed nets, water treatment, health insurance, or cross-country longevity/health-welfare accounting), *pollution* (air or water pollution, emissions, or climate change), or *other* (everything else, e.g. education, cash transfers, transportation safety, or general cross-country welfare accounting).

.3 Model and tooling

Classification was performed using Claude Sonnet 5 (Anthropic; model identifier `claude-sonnet-5`), accessed through Claude Code. Rather than a single long-running conversation, each paper was classified by an independent model subagent given only that paper's file path, the relevant classification rubric, and instructions to search the paper's full text (using a full-text search utility) before reading more broadly if needed, and to justify its answer with direct textual quotations.

.4 Replication: steps and prompts

The exercise was performed using Claude Sonnet 5 (Anthropic; model identifier `claude-sonnet-5`) run inside Claude Code. Each paper was classified by launching a separate Claude Code subagent – via Claude Code's built-in `Agent` tool – rather than continuing a single conversation across papers, so that each classification decision was made independently and without contamination from other papers' context. In outline:

1. **Run Stage 1.** For each paper, launch one Claude Code `Agent` subagent running Claude Sonnet 5 with the Stage 1 prompt template below, in parallel batches of up to 25 papers.
2. **Aggregate.** Collect the returned lines into a running Stage 1 results table.
3. **Filter and run Stage 2.** Restrict to papers coded `use_vsl` = 1 not yet in the Stage 2 table, and repeat classification steps with the Stage 2 prompt template, again one Claude Sonnet 5 subagent per paper.
4. **Spot-check.** Periodically re-verify a sample of negative classifications by hand; if a systematic tooling issue is found, re-audit the affected batches.

Stage 1 prompt template (one Claude Sonnet 5 subagent per paper):

```
Read CLASSIFICATION_INSTRUCTIONS.md fully (hard rule #0: use a full-text
search tool that does not silently skip files -- e.g. ‘command grep‘ or
ripgrep, never a wrapper that may respect .gitignore -- and Stage 1
definitions of estimate_vsl/use_vsl).

Classify this paper for Stage 1.
```

```
Paper: <paper ID> ("<title>")
File: <path to converted text/markdown>

Use ‘command grep -n -i "statistical life\|VSL\b\|DALY\|disability.adjusted
life" <file>‘ first, then Read more broadly if needed.

Respond with EXACTLY ONE line:
<DOI>,<estimate_vsl 0/1>,<use_vsl 0/1>
Then a REASON line with textual evidence.
```

Stage 2 prompt template (one Claude Sonnet 5 subagent per `use_vsl`=1 paper):

```
Read CLASSIFICATION_INSTRUCTIONS.md fully, especially the Stage 2 section
(hard rule #0: use ‘command grep‘, never bare ‘grep‘).

Classify this paper for Stage 2 (already use_vsl=1 in Stage 1).
Paper: <paper ID> ("<title>")
File: <path to converted text/markdown>

Six binary columns: developing_country, nonmarginal_risk, transfer,
transfer_correct, self_aware, givewell. Use the worked examples in the
instructions doc to calibrate ‘transfer‘ especially: [named-funder
transfer case]; [diffuse/collective-funder transfer case]; [rhetorical
calculation that never feeds a real decision, transfer=0]; [purely
domestic or purely descriptive academic exercise, transfer=0].

Use ‘command grep -n -i -B5 -A10 "statistical life\|VSL\b\|DALY\|fund\|
financ\|polic" <file>‘ first, then Read more broadly if needed.

Respond with EXACTLY ONE line:
<DOI>,<developing_country>,<nonmarginal_risk>,<transfer>,
<transfer_correct>,<self_aware>,<givewell>
Then brief REASON lines with textual evidence for each column.
```

.5 Full classification results

Table 1 reports the Stage 2 classification of all 240 papers coded `use_vsl` = 1. `Category` is reported only for papers with `Developing Country` = Y. The papers are identified by NBER working paper number.

Table 1: Stage 2 classification of all NBER working papers coded `use_vsl` = 1 (n = 240)

Paper	Dev. Country	Non-marg. Risk	Transfer	Transfer Correct	Self-Aware	GiveWell	Category
w7716	Y	–	Y	–	–	–	Health
w7717	Y	–	Y	–	–	–	Health
w7760	–	Y	–	–	Y	–	–
w8481	Y	Y	Y	–	–	–	Health
w8755	–	–	–	–	–	–	–
w8818	–	Y	–	–	–	–	–
w9396	–	Y	–	–	–	–	–
w9487	–	–	–	–	–	–	–
w9765	Y	Y	–	–	–	–	Health
w10053	–	–	–	–	–	–	–
w10328	–	Y	–	–	–	–	–
w10511	Y	Y	–	–	–	–	Health
w10737	–	Y	–	–	–	–	–
w10824	–	–	–	–	–	–	–
w10906	–	–	–	–	–	–	–
w11288	Y	–	Y	–	–	–	Health
w11316	–	–	–	–	–	–	–
w11405	–	Y	–	–	–	–	–
w11934	–	–	–	–	–	–	–
w11956	–	–	–	–	–	–	–
w12054	–	Y	–	–	Y	–	–
w12092	–	Y	–	–	–	–	–
w12122	Y	Y	–	–	–	–	Other
w12184	–	Y	–	–	–	–	–
w12294	–	–	–	–	–	–	–
w12836	–	Y	–	–	Y	–	–
w13087	Y	Y	–	–	–	–	Health
w13178	–	–	–	–	–	–	–
w13396	Y	Y	Y	–	Y	–	Health
w13490	–	Y	–	–	Y	–	–
w13599	–	Y	–	–	Y	–	–
w13877	–	–	–	–	–	–	–
w14011	Y	Y	–	–	–	–	Pollution
w14093	Y	Y	–	–	Y	–	Health
w14504	–	–	–	–	–	–	–
w14522	–	Y	–	–	–	–	–
w14823	–	Y	–	–	–	–	–
w14977	–	–	–	–	–	–	–
w15174	–	Y	–	–	–	–	–
w15235	–	Y	–	–	–	–	–

Table 1 continued (page 2 of 6)

Paper	Dev. Country	Non-marg. Risk	Transfer	Transfer Correct	Self-Aware	GiveWell	Category
w15280	Y	–	–	–	–	–	Health
w15383	–	–	–	–	–	–	–
w15471	–	Y	–	–	Y	–	–
w15574	–	Y	–	–	–	–	–
w15649	–	Y	–	–	Y	–	–
w15658	–	–	–	–	–	–	–
w15666	–	–	–	–	–	–	–
w16108	–	Y	–	–	–	–	–
w16352	Y	Y	–	–	Y	–	Other
w16599	Y	Y	–	–	Y	–	Other
w16722	–	–	–	–	–	–	–
w16844	–	–	–	–	–	–	–
w16953	–	Y	–	–	–	–	–
w17094	–	Y	–	–	–	–	–
w17170	–	–	–	–	–	–	–
w17937	–	Y	–	–	–	–	–
w18267	–	–	–	–	–	–	–
w18468	–	Y	–	–	Y	–	–
w18815	–	–	–	–	–	–	–
w18849	–	–	–	–	–	–	–
w19058	–	–	–	–	–	–	–
w19494	Y	–	–	–	–	–	Other
w19881	–	–	–	–	–	–	–
w19956	–	Y	–	–	–	–	–
w20215	–	Y	–	–	Y	–	–
w20330	–	Y	–	–	–	–	–
w20460	Y	–	–	–	–	–	Pollution
w20500	–	–	–	–	–	–	–
w20525	–	–	–	–	–	–	–
w20653	–	Y	–	–	–	–	–
w20810	–	–	–	–	–	–	–
w21129	–	–	–	–	–	–	–
w21291	–	–	–	–	–	–	–
w21308	–	Y	–	–	Y	–	–
w21324	Y	Y	Y	–	–	–	Health
w21383	–	–	–	–	–	–	–
w21635	–	Y	–	–	–	–	–
w21671	–	–	–	–	–	–	–
w22092	–	–	–	–	–	–	–
w22105	Y	–	–	–	–	–	Pollution

Table 1 continued (page 3 of 6)

Paper	Dev. Country	Non-marg. Risk	Transfer	Transfer Correct	Self-Aware	GiveWell	Category
w22155	–	–	–	–	–	–	–
w22261	–	–	–	–	–	–	–
w22579	–	–	–	–	–	–	–
w22610	–	–	–	–	–	–	–
w22769	–	–	–	–	–	–	–
w22796	–	–	–	–	–	–	–
w22814	–	–	–	–	–	–	–
w22899	–	–	–	–	–	–	–
w23340	–	–	–	–	–	–	–
w23346	–	Y	–	–	Y	–	–
w23386	–	–	–	–	–	–	–
w23413	–	–	–	–	–	–	–
w23417	–	–	–	–	–	–	–
w23791	–	–	–	–	–	–	–
w23852	–	–	–	–	–	–	–
w23866	–	–	–	–	–	–	–
w23873	–	–	–	–	–	–	–
w23966	–	–	–	–	–	–	–
w24183	Y	–	–	–	–	–	Pollution
w24206	Y	Y	Y	–	–	–	Health
w24247	Y	Y	–	–	Y	–	Health
w24308	–	–	–	–	–	–	–
w24607	–	Y	–	–	–	–	–
w24688	Y	–	–	–	–	–	Pollution
w24802	–	Y	–	–	–	–	–
w25055	–	Y	–	–	–	–	–
w25241	–	–	–	–	–	–	–
w25339	–	–	–	–	–	–	–
w25641	–	–	–	–	–	–	–
w25661	–	–	–	–	–	–	–
w25731	Y	–	Y	–	–	–	Health
w25791	–	–	–	–	–	–	–
w25910	–	–	–	–	–	–	–
w25971	–	Y	–	–	–	–	–
w26068	–	Y	–	–	Y	–	–
w26381	–	–	–	–	–	–	–
w26775	Y	Y	Y	–	–	–	Health
w26783	–	–	–	–	–	–	–
w26804	–	–	–	–	–	–	–
w26882	–	Y	–	–	–	–	–

Table 1 continued (page 4 of 6)

Paper	Dev. Country	Non-marg. Risk	Transfer	Transfer Correct	Self-Aware	GiveWell	Category
w26942	–	Y	–	–	–	–	–
w26981	–	Y	–	–	Y	–	–
w26984	–	Y	–	–	–	–	–
w27009	–	Y	–	–	–	–	–
w27046	–	Y	–	–	–	–	–
w27059	–	Y	–	–	–	–	–
w27062	–	Y	–	–	–	–	–
w27104	–	Y	–	–	–	–	–
w27121	–	Y	–	–	Y	–	–
w27135	–	–	–	–	–	–	–
w27172	–	–	–	–	–	–	–
w27245	–	–	–	–	–	–	–
w27285	–	–	–	–	–	–	–
w27340	–	Y	–	–	–	–	–
w27398	Y	–	–	–	–	–	Pollution
w27430	–	–	–	–	–	–	–
w27458	–	Y	–	–	–	–	–
w27578	–	–	–	–	–	–	–
w27599	Y	–	Y	–	–	–	Pollution
w27757	Y	Y	–	–	Y	–	Health
w27794	–	Y	–	–	–	–	–
w27848	–	Y	–	–	–	–	–
w27863	–	Y	–	–	Y	–	–
w27933	–	–	–	–	–	–	–
w28031	–	Y	–	–	–	–	–
w28064	–	Y	–	–	–	–	–
w28071	–	–	–	–	–	–	–
w28199	–	–	–	–	–	–	–
w28202	–	–	–	–	–	–	–
w28282	–	Y	–	–	–	–	–
w28492	Y	Y	Y	–	–	–	Health
w28550	–	–	–	–	–	–	–
w28601	–	Y	–	–	Y	–	–
w28617	–	–	–	–	–	–	–
w28619	–	–	–	–	–	–	–
w28622	–	–	–	–	–	–	–
w28636	–	Y	–	–	Y	–	–
w28707	–	–	–	–	–	–	–
w28708	Y	–	–	–	–	–	Pollution
w28734	Y	–	Y	–	–	Y	Health

Table 1 continued (page 5 of 6)

Paper	Dev. Country	Non-marg. Risk	Transfer	Transfer Correct	Self-Aware	GiveWell	Category
w28848	–	–	–	–	–	–	–
w28867	–	Y	–	–	–	–	–
w29063	–	Y	–	–	–	–	–
w29071	–	–	–	–	–	–	–
w29104	–	Y	–	–	–	–	–
w29152	–	–	–	–	–	–	–
w29419	–	–	–	–	–	–	–
w29454	–	–	–	–	–	–	–
w29465	–	–	–	–	–	–	–
w29574	–	Y	–	–	–	–	–
w29682	Y	Y	–	–	–	–	Health
w29718	–	Y	–	–	Y	–	–
w29764	–	Y	–	–	Y	–	–
w29809	–	Y	–	–	–	–	–
w29845	–	–	–	–	–	–	–
w29854	–	–	–	–	–	–	–
w29884	–	–	–	–	–	–	–
w30104	–	Y	–	–	–	–	–
w30118	–	–	–	–	–	–	–
w30181	–	–	–	–	–	–	–
w30286	–	–	–	–	–	–	–
w30303	–	Y	–	–	–	–	–
w30391	Y	–	Y	–	–	–	Health
w30553	–	–	–	–	–	–	–
w30565	Y	Y	–	–	–	–	Health
w30619	Y	Y	Y	–	–	–	Health
w30626	–	–	–	–	–	–	–
w30648	Y	–	–	–	–	–	Pollution
w30663	–	Y	–	–	–	–	–
w30670	–	–	–	–	–	–	–
w30702	–	–	–	–	–	–	–
w30801	–	–	–	–	–	–	–
w30830	–	–	–	–	–	–	–
w30852	–	Y	–	Y	Y	–	–
w30966	–	–	–	–	–	–	–
w31139	–	–	–	–	–	–	–
w31162	Y	–	Y	–	–	Y	Health
w31203	Y	–	–	–	–	–	Health
w31361	–	–	–	–	–	–	–
w31379	–	–	–	–	–	–	–

Table 1 continued (page 6 of 6)

Paper	Dev. Country	Non-marg. Risk	Transfer	Transfer Correct	Self-Aware	GiveWell	Category
w31423	Y	Y	–	–	–	–	Health
w31475	–	–	–	–	–	–	–
w31555	–	Y	–	–	–	–	–
w31593	–	–	–	–	–	–	–
w31758	–	Y	–	–	Y	–	–
w31809	Y	–	–	–	–	–	Pollution
w31812	Y	Y	Y	–	Y	–	Health
w31885	–	–	–	–	–	–	–
w31957	–	Y	–	–	–	–	–
w31999	Y	Y	–	–	Y	–	Other
w32035	–	–	–	–	–	–	–
w32059	–	Y	–	–	–	–	–
w32110	–	–	–	–	–	–	–
w32195	–	–	–	–	–	–	–
w32235	–	–	–	–	–	–	–
w32349	–	Y	–	–	Y	–	–
w32548	–	–	–	–	–	–	–
w32558	Y	Y	–	–	–	–	Health
w32613	–	–	–	–	–	–	–
w32728	Y	–	Y	–	–	–	Pollution
w32742	Y	Y	Y	–	–	–	Other
w32831	–	Y	–	–	–	–	–
w32870	–	Y	–	–	–	–	–
w32955	–	–	–	–	–	–	–
w32984	Y	Y	–	–	–	–	Health
w33032	–	–	–	–	–	–	–
w33143	Y	–	–	–	–	–	Pollution
w33249	–	Y	–	–	–	–	–
w33279	–	Y	–	–	–	–	–
w33429	–	Y	–	–	Y	–	–
w33447	–	–	–	–	–	–	–
w33557	Y	–	–	–	–	–	Health
w33602	–	Y	–	–	–	–	–
w33699	–	Y	–	–	–	–	–
w33905	–	Y	–	–	–	–	–
w34081	Y	Y	–	–	Y	–	Pollution
w34152	Y	Y	Y	–	Y	Y	Other
w34339	Y	–	Y	–	–	–	Other
w34357	Y	–	Y	–	–	–	Pollution
w34525	–	–	–	–	–	–	–

Notes: Dev. Country = paper applies VSL/DALYs to a low- or middle-income population. Non-marg. Risk = VSL applied to a non-marginal risk. Transfer = population priced at its own VSL for a decision financed/decided by a different, wealthier entity. Transfer Correct = funder's own VSL applied to value an externally financed population. Self-Aware = authors acknowledge the application is imperfect (defined only when Non-marg. Risk or Transfer equals

Y). GiveWell = intervention evaluated was GiveWell-funded. Category = subject-matter classification (Health, Pollution, or Other), populated only when Dev. Country = Y.